UNIT 1 · KINEMATICS · WEEK 4

Problem Set 6 — Free Fall & Projectiles

Physics 2026–27 · due at the start of class, Monday, October 12 · Unit 1 Test Thu, Oct 15

NAME ____________________ DATE ____________

How to work this set. Convention for the whole set: **up = positive**, so $g = -9.8\ \text{m/s}^2$ everywhere. For projectiles, run the three-step plan from class: *decompose, vertical KDEA for t, horizontal KDEA for Δx*. Horizontal and vertical never share an equation — time is the only bridge. Use ≈ honestly when the numbers don't land clean.

1 At the peak

CONCEPT

a. A ball thrown straight up is momentarily at rest at the top of its flight. What is its acceleration at that instant? (A number, a sign, and one sentence.)

b. On the Moon, an astronaut drops a feather and a hammer together. What happens, and why doesn't the same thing appear to happen on Earth?

2 Off the bridge

FREE FALL

A rock is dropped (released from rest) off the Rickenbacker bridge and hits the water $2.2\ \text{s}$ later.

a. How high above the water was the release point?

b. How fast was the rock moving at impact? Report the speed and the direction.

K	EVERY GIVEN, WITH UNITS · $G = -9.8\ \text{M/S}^2$
D	THE TARGET & ITS UNIT
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

3 Straight up

FREE FALL

A tennis ball is thrown straight up at 15 m/s from ground level. Two targets, two KDEA passes — start K from scratch each time:

a. How high does it rise? (What do you know about v_f at the peak?)

b. How long is it in the air, launch to landing? (What do you know about Δy for the full trip?)

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

4 Decompose only

COMPONENTS

A ball is launched at 24 m/s, 35° above the horizontal. Find v_x and v_{iy} — the Workshop III move, nothing more. Then one sentence: which of the two values will still be true at the very end of the flight, and why?

WORK

5 Off the shelf

HORIZONTAL LAUNCH

A ball rolls off a shelf 1.8 m above the floor, moving horizontally at 3.5 m/s. (A horizontal launch means $v_{iy} = 0$ — the decomposition is done for you.)

a. Vertical pass: how long is it in the air?

b. Horizontal pass: how far from the base of the shelf does it land?

K	VERTICAL KNOWNS, THEN HORIZONTAL KNOWNS
D	TWO TARGETS, TWO UNITS
E	VERTICAL: KINEMATICS · HORIZONTAL: $\Delta X = V_x T$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

6 The corner kick

FULL PROJECTILE

A soccer ball is kicked from ground level at `21 m/s`, `40°` above the horizontal, and lands back at ground level. Run the full three-step plan: decompose, vertical for time of flight ($\Delta y = 0$), horizontal for range.

K	COMPONENTS FIRST — THEY FEED K
D	THE TARGETS & THEIR UNITS
E	ONE EQUATION PER DIRECTION — NEVER MIXED
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

7 Two coins

INDEPENDENCE

Two identical coins leave the edge of a table at the same instant and from the same height: one is simply *dropped*, the other is flicked *horizontally* at 2 m/s. Which coin hits the floor first? Explain using the independence of horizontal and vertical motion — this is the single idea the whole week hangs on.

ANSWER

*Graded on the homework 4–3–2–1 scale. This is the last problem set before the **Unit 1 Test (Thu, Oct 15)** — every problem type here reappears there.*